For Cisgender Ears: Presenting Transgender Identities Through the Voice

---

A Thesis
Presented to
The Division of Philosophy, Religion, Psychology, and Linguistics
Reed College

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Arts

Wren Arbor

May 2026

Approved for the Division
(Linguistics)

_____

Allison Casar

Acknowledgements

I want to thank a few people.

List of Abbreviations

CoG Center of Gravity; a measure in Hz of mean acoustic pitch at a given moment, weighted by amplitude.

Table of Contents

Introduction 1

Chapter 1: Literature Review 5
1.1 Theoretical Motivations and Background 5
1.2 The Sociophonetics of /s/ 11
1.3 extra stuff 19

Chapter 2: Methodology 21
2.1 Participants 21
2.1.1 Survey 21
2.2 Conversations 22
2.3 Interviews 23
2.4 Analytical Approach 23

Chapter 3: Participant Profiles 25

Chapter 4: Quantitative Results and Analysis 27
4.1 Within-speaker 27
4.2 Between-speaker 27

Chapter 5: Discussion and Qualitative Analysis 29
5.1 Comparison with Prior Results 29

Conclusion 31
5.2 Directions for Future Research 31

Appendix A: Survey 33

Appendix B: Interview Questions 35

List of Tables

2.1 Demographic Survey Responses [this table will be filled in once data has been collected.] 22
2.2 Gender Background Responses (HRT: hormone replacement therapy, VFT: vocal feminization therapy: VFS: vocal feminization surgery) [this table will be filled in once data has been collected.] 22

List of Figures

1.1 Spectrum of mean /s/ CoG by identity group from Podesva and Van Hofwegen (2015, p. 181), synthesizing with data from Hazenberg (2012) and Zimman (2013). [TODO: recreate this visual so that it's easier to read, all in one single spectrum, and has all of hazenberg's data rather than just the women part of it!!] 12
1.2 Diagram from Calder (2019a, p. 340) illustrating indexical orders of /s/ frontness and corresponding fractal recursive social oppositions across two San Fransisco drag communities. 17

Abstract

[The abstract will go here]

Introduction

Along with the visual presentation of one's body, the sound of one's voice is among the most strongly gendered aspects of behavior. Trans people[1] know this quite intimately, as our speech is highly scrutinized by those who – consciously or unconsciously – seek to enforce the gender binary (Jessee and Calder 2025). In this thesis, I investigate how this knowledge manifests by comparing English-speaking transfeminine[2] speakers' linguistic behavior across two conversational contexts: a conversation with a cis[3] woman, and a conversation with a trans woman. I focus on their pronunciation of the /s/ sound (the first sound in 'safe' and the last sound in 'face'), a well-studied site of gendered linguistic variation.

It has become widely accepted both in the academic study of gender and parts of the popular conciousness that gender is performative (Butler 1990; Henry and Hall 2025). Scholars of sociolinguistics and linguistic anthropology have come to see semiotic (or meaning-making) processes as a primary mechanism of performativity, by which actions and behaviors are sites of meaning-making, and these meaning-making acts constitute performative enactments of gender (Henry and Hall 2025). A primary mechanism of meaning-making is indexicality, by which a perceiver construes a particular phenomenon to be a sign of some meaning by virtue of their belief in and/or knowledge of a co-occurence relationship between the phenomenon and the meaning (Gal and Irvine 2019). Variation in language use is indexical; listeners believe that certain (types of) people use certain linguistic forms (co-occurence), and that the use of a particular linguistic form means something about the speaker who used it. Particular indexicalities may come to be entrenched as expectations and norms, deeming the voices of marginalized speakers, who may not make use of these indexicalities in their own communities or beliefs, inadequate and inappropriate (eg. Inoue 2006; Inoue 2003; Flores and Rosa 2015; Heffernan 2023).

This raises the point that linguistic forms are only interpretable as signs within

[1] I use 'transgender' or 'trans' to refer to those who identify with this label. Many transgender people understand the term broadly to include all who do not fully identify with the gender assigned to them at birth. However, the term is sometimes narrowed to refer mostly to transgender men and transgender women, who identify with the opposite binary gender of the gender assigned to them at birth.

[2] I use the term 'transfeminine' to those who identify with this label, often feminine nonbinary people (especially those assigned male at birth, for whom femininity would be more of a transition) and trans women.

[3] I use 'cisgender' or 'cis' to refer to people who are not transgender.

a set of culturally-available indexical expectations (Calder 2021) that a listener might apply in perception of a speaker's voice. Existing studies primarily focus on only one set of these expectations, either the powerful, majoritarian set among the cis straight white population (eg. Hazenberg 2012; Podesva and Van Hofwegen 2015), or the alternative sets constructed within local, minoritarian, marginalized communities (eg. Mendoza 2024; Calder 2019a). Within the realm of the study of trans speech, the former type of study often interprets trans voices in their deviance from or compliance with cisnormativity, and runs the risk of tacitly enforcing the norms of the gender binary, marginalizing anti-normative trans voice practices (Calder and A. Steele 2025). Furthermore, focus within a single type of theoretical listener – hegemonic or otherwise – often sidelines the intersubjective nature of identity, by which identity emerges in the establishment and contrast of self and other, speaker and listener (Bucholtz and Hall 2010)– especially the fact that all speakers will be subjected to powerful, majoritarian norms. Recent research has taken steps to place more even focus on both sides of the speaker-listener pair, investigating how trans voices are perceived by listeners (Jessee and Calder 2025) and trans people's understandings of these perceptions (Heffernan 2023). However, research which treats the listener as a variable, investigating how trans speakers' understandings of how they will be perceived manifest in their vocal practice, has not yet been conducted. By filling this gap, I hope to demonstrate the diverse strategies trans speakers employ to resist transphobic perceptions.

Language and gender research has found that the pronunciation of the /s/ sound is subject to Western hegemonic white cisheteronormative gender norms; higher-pitched /s/ pronunciations are expected of and associated with feminity (see Calder 2020). Jessee and Calder (2025) find that cisgender listeners impose this standard more strictly on trans voices than on cis voices, expecting /s/ pitch to be higher from trans women than cis women and lower from trans men than cis men. Although transfeminine people may not be consciously aware of /s/ pitch as an index of femininity, they nonetheless have been shown to pronounce /s/ more closely to cis women than to men (eg. Hazenberg 2012), and to understand that cis listeners apply strict gender norms to their voices, and have strategies to navigate these expectations. One of the transfeminine speakers in Heffernan (2023) noted that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly", and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). Due to the theoretical richness of research already conducted on /s/ pitch, I investigate whether variation in /s/ pitch is involved in the sort of modulation Heffernan's participant mentions.

In order to investigate how trans speakers' understandings of how their voices are perceived manifest in their vocal practice, I audio-recorded four transfeminine speakers, each of which had one conversation with a cis stranger, and one conversation with a trans stranger. I compare each speakers' /s/ pitch between the two interactions. I predict that one possible outcome would be in line with the experience of Heffernan's speaker, where speakers raise their /s/ pitch to accomodate the demands of cisnormativity and avoid the reprocussions of choosing not to do

so. However, it is likely that some transfeminine people feel differently than the speaker in Heffernan, instead choosing not to raise /s/ pitch as a move of resistance or to present a uniquely trans feminine identity beyond cis norms. For these reasons, I hypothesize that trans speakers will employ diverse vocal strategies which may help them defend their self-determined identities, avoid misgendering, resist gender norms, and forge new gender possibilites.

Chapter 1

Literature Review

1.1 Theoretical Motivations and Background

At the broadest level, this thesis seeks to contribute to an understanding of gender from a trans and queer critical perspective. Following the revolutionary work of Judith Butler (1990), gender scholarship has come to regard gender as performative. Most simply, this means that gender does not exist outside of culture and society, and is constantly constructed and reconstructed by social behavior. Linguistic behavior has always played a major role in performativity; Butler borrowed the term from linguist J. L Austin (1962), who noticed that by using language we do not simply make statements about the world, we affect and shape it. Butler adapted and revised the theory, redeveloping it on a sophisticated social-theoretical background (Henry and Hall 2025). This renewed the interest of sociolinguists and linguistic anthropologists in performativity, especially those in the study of language and gender (Henry and Hall 2025; Bucholtz 2014). These scholars would come to connect performativity with the sophisticated theorizations of the social and cultural dynamics of language use they had already developed, primarily their models of semiotics (meaning-making) and indexicality (Henry and Hall 2025). For this thesis, I adopt the model of semiotics developed by Gal and Irvine (2019).

In Gal and Irvine's (2019) model, individuals make meaning of their world through conjectures. Drawing upon their knowledge and understanding of the world, they recognize a phenomenon and conjecture that it may be a sign, i.e. it may be meaningful. They conjecture that the sign signals (means or represents) a particular object, its meaning. Essential to note here is that these conjectures constitute the sign-object relationship; nothing is a sign in itself, independent of a perceiver conjecturing that it is a sign and it signals an object (Gal and Irvine 2019, p. 89). Also important is that these conjectures are based on the perceiver's knowledge and understanding of the world, and which are constructed and mediated by their cultures, communities, and individual social identities– their lived experience (Gal and Irvine 2019; see also Calder 2021). Part of a conjecture is its "grounds": the reasoning the conjecture provides as to why its phenomenon is a sign and why it signals this object. A conjecture is considered to be indexical if

the grounds of its claim that a phenomenon is a sign of a particular object are a knowledge or belief that the sign often co-occurs with, points to, or is caused by its object (Gal and Irvine 2019, p. 96).

Sociolinguistics often focuses on finding correlations between differences in language use (eg. pronunciation, grammar, word choice) and differences in social identity, persona, or context (Eckert 2012). Particular linguistic forms are then said to index the social identities of those who use them, on the grounds of this relationship of co-occurence (Eckert 2012). Termed by Silverstein (2023, p. 17) the "social-indexical performativity of language-in-use", this semiotic process has social performative effects, and is at the core of social performativity, including gender performativity (Henry and Hall 2025). If a speaker uses a particular linguistic form, and the listener believes that form to be commonly used by those of a particular gender identity and thus conjectures that linguistic behavior as indexing that speaker's gender, this socioculturally-mediated perceiving act has performatively constructed the speaker's gender. That there are two actors in play here, a speaking subject and a listening subject, led Bucholtz and Hall (2005; 2004; 2010) to understand the performative construction of gender (and identity categories more broadly) as an intersubjective process.

Bucholtz and Hall characterize this intersubjective process as a negotiation between the indexical goals of the speaker and the indexical goals of the listener (eg. Bucholtz and Hall 2005, p. 606). Although it is possible that a listener may construe an indexical meaning from a linguistic form used by a speaker who has no knowledge of, or intent to make use of, that form's indexicality, agentivity on the part of the speaker is common. Speakers have interactional goals, and hope that their behavior will be construed as indexical of those goals, whether it be related to identity, eg. presenting feminine, or other aspects such as persona and stance, eg. seeming like a 'diva' (Podesva 2007). This agentic view arose principally in reaction to earlier models which stripped speakers of agency and individuality, instead understanding their linguistic behavior only to index their static, essentialized identities (Eckert 2012). In practice, this turn reframes "she speaks this way because she is a woman" as "she chooses this form of speech to construct a feminine identity." Acknowledging that individuals have choices about how to present themself and granting them authority over their self-identification is also a move towards alignment with queer and trans understandings of identity; trans people in the US often consider gender identity to be self-determined (see Zimman 2017b), rather than a consequence of bodily characteristics. Despite the value of emphasizing the speaker's goals to legitimize their identities and agency, it is nonetheless the case that the listener is the powerful other side of this negotiation.

Inoue (2006; 2003) argues that in cases where the listener occupies a privileged social position of power over the speaker, they may inhabit a priviliged 'listening subject' position in which the beliefs required to uphold this social hierarchy may play a primary role in their construal of that speaker's indexical behavior and override or erase the speaker's intentions. Studying the emergence of the notion of 'women's language' in Japan, Inoue finds that to Meiji period elite male intellectuals, schoolgirls inhabited a social position which represented a modernizing

breach from traditional Japanese social structure, and a threat to their power. For this reason, when they overheard schoolgirls using the verb ending forms *teyo* and *dawa*, they construed these endings as indexical of degeneracy and social decline. Of course, schoolgirls themselves did not intend to use these forms to index a 'degenerate' identity; the purported degeneracy these forms were said to represent was a social fabrication rather than a truth about any actions these girls were involved with. In this way, male intellectual overhearers constructed a listening subject position which made negative judgemental conjectures about the indexicality of schoolgirls' speech, and constructed a fictional object for the these speech forms to signal. The male listening subject sets the narrative, erasing schoolgirl speakers' semiotic agency.

Following this important insight, research has extended theorization of the listening subject to consider race and queerness, researching white (Flores and Rosa 2015) and cisgender listening subjects (Calder 2021; Heffernan 2023; Jessee and Calder 2025; Calder and A. Steele 2025). Flores and Rosa (2015) find that when white listeners racialize speakers as non-white, they justify their position of power over those speakers by construing those speakers' speech as indexing an underspecified 'inappropriateness' for the situation (eg., the classroom), regardless of those speakers' adherence to linguistic norms for standard English, constructing a white-supremacist white listening subject which will always deem non-white speech deficient and indexical of non-white inferiority. Jessee and Calder (2025) find that cisgender listeners apply stricter gendered listening standards to an audio clip of a voice when they are told the speaker of the clip is a trans woman than when told the speaker is a cis woman, similarly situating sufficient gender performance out of reach for transgender speakers in a move to deny their identities.

Drawing on these concepts of white, cis, and male listening subjects, Calder (2021) and Calder and A. Steele (2025) raise an important critique of the sociolinguistic project, questioning which listening subjects researchers themselves may be inhabiting in their interpretation of speakers' language use. Following the conjecturing model of Gal and Irvine (2019), they note that researchers collect numerical correlation data between linguistic forms and social aspects, and use this as the knowledge on which to base their conjectures of indexical links. Incorporating the research on the listening subject, they conclude that although this data may seem objective, empirical, and scientific, purely describing the world as it is (see also Zimman 2021a), listening practices inevitably are biased by the socio-cultural positioning of the researcher and the scholarship that researcher builds upon. This critique raises the question of how to research the social-indexical performativity of language-in-use without taking up and reinforcing violent listening practices and marginalizing speakers who are not white, cis, heterosexual men. Work which addresses this question employs advanced models of indexicality which encourage sociolinguistic study to move from simply asking what meanings participants index to investigating the socio-cultural structure of individuals' indexical knowledge, and the intersubjective process of indexicality (Calder 2021; Silverstein 2003; Silverstein 2023; Barrett and Hall 2024). The concepts of the indexical field as adapted by Calder (2021), and indexical (mis)recognition as de-

veloped by Barrett and Hall (2024) have been used to model indexical knowledge and the relativity and intersubjectivity of indexicality.

The concept of the indexical field, introduced by Eckert (2008), has been adopted by Calder (2021) to capture the way in which a particular phenomenon may be conjectured to have different meanings by different groups of people. The concept of the indexical field emphasizes, connecting to Gal and Irvine's (2019) semiotic model, that indexical relationships are conjectured through listeners' knowledge. In this way, a particular linguistic form may be construed as a sign of any number of meanings, depending on which knowledge the listener uses to conjecture the relationship, and what conclusion they come to in conjecturing. Although this knowledge is mediated and constructed differently by each individual's social and cultural lived experience, it is nonetheless to a large extent shared. For example, Inoue's (2003) listening subject was not a single Meiji-era Japanese male intellectual, it was an ideological position constructed and inhabited by this male intellectual social class. Individuals who inhabit similar social and cultural backgrounds and social positions, as well as co-members of communities, are likely to draw upon a similar base of knowledge, and are likely to consider certain phenomena signs with a particular set of possible meanings. Calder (2021) conceptualizes the indexical field as this set of meanings which may be shared by a set of people with shared epistemology. By making explicit whose indexical field researchers are investigating and interpreting signs through, the fact that this interpretation is coming from particular perspectives is forefronted, preventing the universalization of the hegemonic indexical field (the white, male, heterosexual, cisgender, middle class, etc. indexical field).

While this makes progress towards a more critical and less biased move to interpreting signs, and much research has been productively done to investigate the indexical fields within particular local communities, especially community-specific systems of categorization and meaning, it is nonetheless untenable to simply split the world into non-overlapping spheres of shared knowledge and shared indexicality. In other words, indexical fields overlap and individuals always exist in multiple spaces, so an awareness of the ways in which indexical conjectures may be made about an individual's behavior must span all indexical fields perceivers may apply to them. The question that arises is: what happens when a speaker and listener do not share an indexical field?

Barrett and Hall (2024) capture this with their concepts of the indexical inventory and indexical (mis)alignment/(mis)recognition. For them, an individual's indexical knowledge is structured in an indexical inventory, which is "an individual's full set of indexical fields." (p. 132) When individuals interact and their indexical inventories share similar indexical fields, they are said to be in indexical alignment. In this case, if a speaker uses a linguistic form that they know to hold a particular indexical meaning within one of the indexical fields in their inventory, their listener would recognize (in Barrett and Hall's term) their indexical behavior as coming from the indexical field on which they are aligned, interpreting it in line with the speaker's intentions. However, in cases of indexical misalignment, the speaker employs a linguistic form with the intent to index a particular meaning

present in the indexical field for that form in their inventory, but the listener does not have that indexical field present in their inventory; there are two possibilities at this point. If the listener has no indexical field in their inventory for that form, they may simply not conjecture about it, not knowing that it was intended to be a sign. If they do have an indexial field for that form in their inventory, but it is not the same as the one the speaker intended to call upon, they may conjecture that the sign means something other than what the speaker had intended.

They cite the "Friend of Dorothy" social routine, documented primarly among white cisgender North American gay communities in the 1980s and 90s, as an example of a sign which makes purposeful use of indexical misalignment. Within this routine, a person asks "are you a friend of Dorothy?" as a coded form of the question "are you gay?", drawing on an in-group indexical field specific to the gay communities in which the routine circulates. The usefulness of the question comes from the exclusivity of its indexical field; asking a straight person whether they are gay straightforwardly may result in violence, but if it is assumed that straight people are unaware of the routine, but that gay people are aware, then it will only be recognized by gay people, removing the risk (Barrett and Hall 2024, p. 129-30). This model complicates the relationship between the speaker and the listener in indexicality significantly, and raises the primary theoretical question this thesis aims to shed insight on: how do speakers model the indexical inventory of those who may be listening to them, and how do these models influence their behavior to help them achieve their indexical goals, whether those goals be alignment and recognition or misalignment and misrecognition?

I argue that this work of modeling the listener's indexicality and adapting to it is not distributed equally, primarily falling to marginalized speakers (especially multiply-marginalized speakers) due to the power dynamics which govern the construction of indexical fields. All speakers are under coercion to understand the white, male, heterosexual, cisgender, middle class hegemonic indexical field, risking negative judgements from the listening subject position associated with that field if they do not learn its norms and store them in their indexical inventories. However, those who inhabit privileged positions are unlikely to face any negative consequences due to ignorance of indexical fields belonging to marginalized groups, as marginalized people do not have the coercive power to construct a judging listening subject position. For this reason, the investigation of marginalized speakers' modeling of and reaction in behavior to listeners who hold privilege over them emerges as the most viable avenue for research on this dynamic generally. It is for this reason, as well as further reasons detailed below, that I choose to focus on transfeminine people for my research.

Heffernan (2023) finds that transfeminine speakers are aware of the cisgender listening subject. When asked questions about how they use their voice and how they feel about their voice, transfeminine speakers in Heffernan's study noted that they fear negative judgements by cis people when linguistic gender norms are not followed. Further, her speakers expressed varied ways of navigating this, from accomodating these expectations by a change in linguistic behavior towards conformance to asserting gender to be self-determined, resisting these judgements,

and talking as one would prefer. One of the transfeminine speakers in Heffernan (2023) noted that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly", and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). However, this speaker, as well as her transfeminine conversation partner, also note that "it's kind of just always incorrect to assume someone's gender", and expressed uncertainty about whether she would want to train her voice to be more hegemonically feminine, as she wants to be able to "[think] about (.) [her] voice ... through uh- like the lens of other trans people", who would not apply the hegemonic indexical field, not assume her gender, and not require this sort of modulation. In this way, the decision whether to shift one's speech in the presence of a privileged listener who may inhabit a judgemental listening subject position is complicated and personal. Experimental investigation of this sort of shifting, comparing transfeminine speakers' indexical behavior when talking to a cisgender person with their indexical behavior when talking to a transgender person has not been conducted.

The work of Becker, Khan, and Zimman (2022), Gratton (2016), and Parnell-Mooney (2019) represent the nearest examples. Becker, Khan, and Zimman (2022) find that trans men and nonbinary people assigned female at birth on testosterone use significantly less creaky voice (known popularly as 'vocal fry') in a reading-passage task compared to in a open-ended conversation. As creaky voice is associated with femininity within the North American English hegemonic indexical field, they tentatively interpret these findings as evidence that in a more scrutinized form of speech (in which, I add, the notion of the cisgender listening subject may be more mentally present), these individuals purposefully move away from normative feminine speech in order to assert their identities as non-women.

This aligns with the findings in a study on a Toronto nonbinary community conducted by Gratton (2016). Gratton finds that variation between *-ing* and *in'* (eg. *working* vs. *workin'*) was more even in queer-safe spaces, and more differentiated between the two in public. This was interpreted as in line with their nonbinary identities, as the *-ing* variant is hegemonically associated with femininity and *in'* with masculinity. In public, speakers avoided the variant more associated with their assigned gender at birth. That cisgender overhearers may be present in the public setting brought the cisgender listening subject's influence to speakers' behavior; the cisgender listener is invested in fitting them into the binary gender they were assigned at birth, so speakers exaggerated their difference from that gender. In queer-safe spaces, the speakers felt free to speak more neutrally and without regard for binary norms. Further, one speaker even noted feeling comfortable making occasional use of a 'manly voice' in these spaces for comedic effect despite the danger of being misgendered as a man as a nonbinary individual assigned male at birth; they knew they would not face misgendering for doing so in the absence of cisgender listeners. Parnell-Mooney (2019) obtains similar findings, seeing that transmasculine speakers and an AMAB nonbinary speaker shift further from the use of a form associated with femininity when interacting with cisgender women than when interacting with cisgender men. Although these studies present important

empirical results, they do not integrate with the theoretical questions of indexical recognition, indexical fields, and the modeling of a listener's indexical knowledge raised above, do not include transfeminine speakers, and do not operationalize as a varaible the trans status of speakers' conversation partners. Filling this gap, I compare transfeminine speakers' indexical behavior across a conversation with a cisgender woman and a transgender woman with reference to their commentaries on their own gender, their voice, and being heard as a transfemine speaker.

For my comparison, I focus specifically on the English /s/ sound [1], as the first sound in 'safe' or the last sound in 'face', due to the richness of existing research on its indexicalities and its use by transgender individuals, especially in connection to the sophisticated theories of indexicality discussed above. The remainder of this section introduces the literature on /s/ and reviews its connections to the concerns of transgender language and indexicality research.

1.2 The Sociophonetics of /s/

By varying the tongue position with which an /s/ is pronounced, a speaker can vary the frequencies (or pitches) of the sound produced; when the tongue position is fronter (closer to the teeth) the resulting /s/ has primarily higher frequencies, when the tongue position is backer (further from the teeth), the resulting /s/ has primarily lower frequencies (Fuchs and Toda 2010). A common measure of this frontness is the spectral center of gravity (CoG), which is the mean of all of the frequencies of an /s/ sound at a given moment in time, weighted for their amplitudes (volumes), measured in Hz. Research on /s/ initially focused on cisgender, heterosexual, white speakers, and found correlations between speakers' binary gender and their /s/ pronunciations, finding women's /s/ pronunciations to be fronter than men's (Flipsen et al. 1999 review 21 such studies). It is important to note that differences in /s/ frontness do not arise from any anatomical differences associated with the sex binary, unlike other gender-correlated sites of variation such as overall vocal pitch; all people with typical vocal tracts have access to roughly the same range of /s/ pronunciations (Zimman 2017c; 2021b).

Early studies which moved to investigate white gay, lesbian, and binary transgender[2] /s/ pronunciations established spectra on which the mean CoG of all /s/ pronunciations of all individuals of the same identity type were placed, as shown in figure 1.1 (Podesva and Van Hofwegen 2015; Hazenberg 2012; Zimman 2013). Across their participants in Redding, California and Ottowa, Ontario, Canada respectively (all of whom were white), Podesva and Van Hofwegen (2015) and Hazenberg (2012) both find that cisgender heterosexual speakers formed the extreme ends of the spectrum, and queer people occupied more central positions.

[1] The slashes surrounding the letter 's' here clarify that I am using the symbol 's' from the International Phonetic Alphabet, representing the voiceless alveolar fricative, rather than referring to the English letter 's' (see International Phonetic Association 1999).

[2] I use 'binary transgender' to refer to transgender people who are not non-binary, identifying primarily with one of the genders of the binary; transgender men and transgender women.

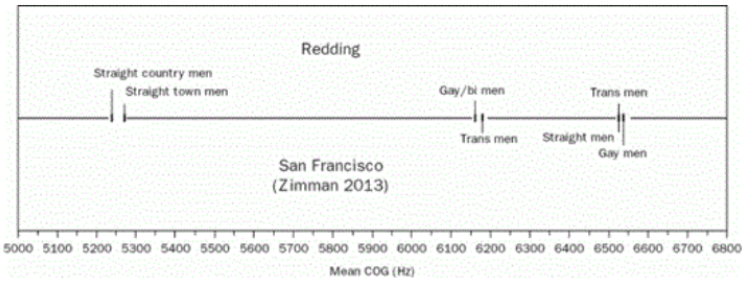


Figure 9.5 Mean center of gravity (Hz) of /s/ by gender identity and sexuality for male-identified speakers in Redding (top), compared to San Franciscans (Zimman 2013) (bottom)

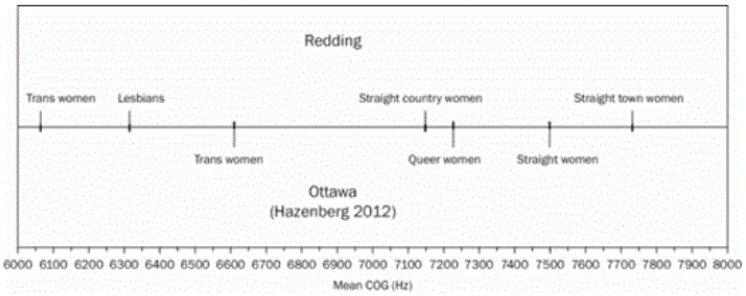


Figure 9.6 Mean center of gravity (Hz) of /s/ by gender identity and sexuality for female-identified speakers in Redding (top), compared to Ottawans (Hazenberg 2012) (bottom)

Figure 1.1: Spectrum of mean /s/ CoG by identity group from Podesva and Van Hofwegen (2015, p. 181), synthesizing with data from Hazenberg (2012) and Zimman (2013). [TODO: recreate this visual so that it's easier to read, all in one single spectrum, and has all of hazenberg's data rather than just the women part of it!!]

In both studies, cisgender queer people represented one step away from the extreme in either direction, with gay men fronting /s/ and queer women backing /s/. The studies differed in their findings on transgender speakers. In Ottowa, trans women occupied the center, with /s/ not significantly different from cis queer women or men, and trans men occupied a similar place as cis queer men, only significantly differing from cis straight people and cis queer women (Hazenberg 2012). In Redding, however, trans women had somewhat backer /s/ and trans men had somewhat fronter /s/ (although numbers were too small for statistical analysis), with trans women more similar to gay men and trans men more similar to lesbians.

Although Podesva and Van Hofwegen (2015), Hazenberg (2012), and Zimman (2013) include white queer cisgender people and white binary transgender people alongside white cisgender heterosexual speakers, Calder (2021) argues these spectra nonetheless paint a picture of cisgender heterosexual white norms. They note that the indexicalities that can be construed from these spectra reflect these norms– /s/ frontness is construed as feminine, and /s/ backness is construed as masculine, since cisgender heterosexual men and women form the extremes of the

/s/ frontness spectrum, and queer speakers are figured as deviant from these indexicalities. Calder (2021) argues that this indexicality makes up the cisgender heterosexual white indexical field for /s/, and the judgements associated with it, by which /s/ pronunciations outside of the white cisgender heterosexual norm are viewed negatively as a failure to properly index one's gender, represent those of the cisgender heterosexual white listening subject.

Research on cisgender, heterosexual white listeners sheds additional light on the structure, application, and boundaries of this indexical field. Strand (1999) finds that when cisgender heterosexual white listeners are asked to categorize audio stimuli on a spectrum of /s/ frontness as either /s/ or /ʃ/ (the English 'sh' sound, as in 'shirt', which is similar to a very backed /s/), they place the boundary higher when told the speaker of the audio clip is a woman than when they are told it is a man, thus indicating an indexical link between /s/ frontness and womanhood/femininity. In men's voices, a number of studies find that fronted /s/ is perceived as gay by cisgender heterosexual white listeners (Pharao et al. 2014; Maegaard and Pharao 2015; Levon 2014; Campbell-Kibler 2011). Munson, Jefferson, and McDonald (2006), replicating Strand's experiment with a larger range of stimuli, find that when listeners perceive a speaker to be lesbian sounding, they place the boundary between /s/ or /ʃ/ lower, thus expecting lesbians to back /s/. These indicate an additional indexicality within the cisgender heterosexual white indexical field of /s/, in which if the combination of /s/ front/backness does not conform with the expectations of the speaker's perceived binary gender, this speaker will be perceived as gay/lesbian.

Jessee and Calder (2025) run another replication of Strand (1999), comparing the placement of the boundary between /s/ and /ʃ/ when the perceiver is told the speaker is trans vs. not told they are trans (and thus is likely to assume that they are cis). They find that unlike in Munson, Jefferson, and McDonald (2006), where queer cisgender speakers are expected not to conform with the expected /s/ pronunciation for their gender, transgender speakers are expected to hyper-conform– trans women are expected to front /s/ more than cis women by all listeners, and listeners who do not know a trans person also expect trans men to back /s/ more than cis men. Although indexicalities certainly would have changed in the 20 years between these studies, it is nonetheless surprising that cisgender heterosexual white listener holds different expectations and perceptions for trans people than for queer cisgender people, especially since it is counter to trans speakers' actual behavior; Podesva and Van Hofwegen (2015) and Hazenberg (2012) both find that trans speakers' /s/ pronunciations are less extremely gender-differentiated than cis speakers'. Jessee and Calder (2025) analyze this through the lens of the cisgender listening subject, seeing that rather than basing perceptions on transgender speakers' actual speech, the cisgender listening subject places strict standards on transgender speakers to conform to the binary more than even cisgender speakers.

By 'conform to the binary', here I mean to conform to the binary gender the trans speaker currently identifies with, rather than the gender that they were assigned at birth. If perceivers conceive of conformance with the binary to be conformance with the gender norms of a trans person's gender assigned at birth (rather than their

actual, current gender), a different understanding emerges in which perceivers expect transgender people to be more deviant from that norm than queer cisgender people of that gender assigned at birth. It is important to note the epistemological disjuncture here; this mode of perceiving transgender speakers requires a rejection of their view of their own gender, in favor of the cis perceiver's view that one's gender is a static consequence of their bodily characteristics. Regardless of whether the perceivers expect trans speech to be highly deviant from their assigned gender at birth or highly conformant with their gender identity, the cisgender listening subject nonetheless construes real trans speakers as inadequately performing gender. Furthermore, this structure of meanings does not include a possibility of indexing a transgender identity using /s/. Trans speakers are 'damned if they do, damned if they don't' – if a trans woman does not front /s/, her transness may be questioned, and she may be heard as a man; if she fronts /s/, she may be construed only as a gay man, not a trans woman. In this way, the cisgender heterosexual white indexical field marginalizes transgender, nonbinary, and other non-cisgender people more than it does queer cisgender people. To avoid applying this marginalizing lens, research that has been conducted on non-cis, non-white speakers' /s/ productions has tended to simply get out of the cisgender listening subject's ears, so to speak, by studying the ways in which these communities forge new indexicalities of /s/ in interactions with other transgender people within local, community-oriented indexical fields.

It is notable that two of the perception studies cited above – Maegaard and Pharao (2015) and Levon (2014) – already provide evidence that the indexical meanings of femininity and gayness for /s/ frontness were specific to the cisgender heterosexual white indexical field, and that other indexical fields may exist. Maegaard and Pharao (2015) finds that listeners in Copenhagen only hear fronted /s/ as feminine and gay when it is within the context of standard Copenhagen Danish speech, and do not hear it as these things when situated in a locally-present speech style known as 'street' speech, as this speech style also happens to feature fronted /s/ and is associated with immigrants, 'gangster'-ness, and masculinity. Although Maegaard and Pharao do not use indexical fields in their analysis, this would nonetheless be an example of a meaning of /s/ within a local indexical field taking precedence over the hegemonic indexical field within a non-white context. Levon (2014) finds that indexicalities within the hegemonic indexical field also are limited to listeners who hold hegemonic attitudes; those who endorse normative masculinity and male gender roles perceived fronter /s/ to be more feminine and gay, whereas those who reject these attitudes do not make these indexical connections. From these studies, it is clear that the construal of fronted /s/ as indexical of femininity and gayness is associated with a white, cisgender, heterosexual, and gender-normative judgemental listening subject, and situated within the corresponding indexical field (as pointed out by Calder 2021), and that other indexicalities and listening subject positions exist. Recent research on trans and non-white speakers' usage of /s/ takes a closer look at these alternative indexical fields and investigates how these indexicalities come about.

Calder (2019a) applies the theories of indexical orders developed by Silverstein

(2023; 2003), and fractal recursivity developed by Irvine and Gal (2000) and Gal and Irvine (1995) in order to investigate this process. As linguistic forms are construed as indexical signs and gain meaning within a particular indexical field, it is possible that new meanings will arise which build off of and are informed by the original meaning. This can take place both within a single indexical field, and cross-cutting the overlaps of multiple. The development of an indexical link between a linguistic form and a social meaning always already emerges informed by an uncountable chain of previous semiotics, and for this reason is said to exist at the nth indexical order (Silverstein 2003). In the case of /s/, Calder understands the indexical link between the gender binary and /s/ frontness to exist at the nth order within the hegemonic indexical field. When cisgender heterosexual white listeners' perception of a speakers' gender does not match with their perception of that speaker's /s/ pronunciation, this recontextualization of /s/ usage forms the $n+1$st indexical order, by which speakers are held to be indexing gender nonconformance. At this point, the epistemological disjuncture mentioned above once again rears its head; Calder's model assumes that cisgender listeners enforce the norms of /s/ based on their perception of the speaker's assigned gender at birth, erasing transgender, nonbinary, and other queer identities and subsuming all gender nonconformance as gay/lesbian. In this way, the $n+1$st order indexicality of /s/ front/backness becomes gay/lesbianness, rather than simply gender nonconformance. It is important to note that although the nth and $n+1$st indexicalities of /s/ are situated at different orders, they nonetheless inform and construct each other, and do not form a chronological order despite their ordered structure.

Within the San Fransisco drag communities Calder (2019a) studies, an additional indexical order of /s/ is forged at the $n+1+1$nd level which is local to these communities' indexical fields, escaping the hegemonic field. A drag community Calder studies in the South of Market (SoMa) neighborhood of San Fransisco figures itself as more radically queer than another drag community they study in the Castro neighborhood. This is reflected by the differences in identities between these communities; SoMa queens generally identify as non-cisgender and challenging to the gender binary, many of them nonbinary or transgender, whereas the queens of the Castro neighborhood primarily identify as cisgender gay men. Calder finds that SoMa queens front /s/ more than Castro queens, and that this forges a new local indexicality to 'fierceness', deviance from gender norms, and and in-group status at the at the $n+1+1$nd level. Their diagram of this structure of meanings is included in figure 1.2. This emerges as an example of indexical orders cross-cutting indexical fields, where meanings in the hegemonic cisgender heterosexual white indexical field came to inform the creation of new meanings in the local, alternative indexical field.

It was mentioned above that within Calder's model, fronting /s/ emerges as deviant from gender norms only in those whose gender is understood as 'male' within the cisgender, transphobic conception of gender as bodily characteristics. It is perhaps surprising, then, that nonbinary and transfeminine speakers in the SoMa drag community, who would likely be gendered this way, front /s/ exactly for the purpose of gender deviance, rather than gender conformance. It is helpful

to introduce a theoretical tool Calder employs to understand the interlinking of identity categories at these different levels of sex at birth, gender identity, sexuality, and trans/queerness– the semiotic process of fractal recursivity, as theorized by Irvine and Gal (2000) and Gal and Irvine (1995).

In fractal recursitivity, a semiotic opposition on one level becomes nested into itself, and this process can happen any number of times (Irvine and Gal 2000; Gal and Irvine 1995). In the first two indexical orders of /s/, both of which are within the cisgender heterosexual hegemonic white indexical field, the gender binary is nested into itself, creating a metaphorical structure where men are split into a binary gender opposition. This can be understood as distinguishing some notion of 'men who are men' from 'men who are women', and 'women who are women' from 'women who are men'. These become the metaphorical representation of gay men and lesbian women, since members of these groups are still fundamentally understood as types of men and types of women, rather than something else entirely, as with transgender and nonbinary people. Is it then the case that by situating the distinction between cisgender gay Castro queens and nonbinary and transgender queer SoMa queens as nesting into the 'men who are women' branch, Calder as the analyst, and possibly the drag queens themselves, tacitly consider these queens as nonetheless a 'type of man/male'? I argue it is not. Instead, in my reading of Calder's work, SoMa queens reinterpret the binary distinction at the $n+1$st level, instead of as gay cisgender men vs. straight cisgender men, as gay men vs. queer non-cisgender people, or as Calder articulates it, gay vs. queer. This binary still fits into the structure, as it is not difficult for SoMa queens to figure gay men as less feminine and less subversive than queer non-cisgender people. From here, the indexicality of /s/ frontness as subversive queer non-cis femininity can be less problematically interpreted.

It is also not necessarily the case that alternative meanings of /s/ emerge through this sort of fractal-recursive process. In some cases, speakers more directly contradict hegemonic meaning structures rather than subversively repurposing and reinterpreting them. For example, A. J. Steele (2022) investigates a community of Columbus, Ohio nonbinary individuals, and finds that not only did speakers' rating of their identification with masculinity not correlate strongly with /s/ backness among white speakers, but that this rating had a positive correlation with /s/ frontness among non-white speakers. This behavior suggests an indexicality within this Columbus nonbinary indexical field which is the inverse of that of the hegemonic indexical field at a higher indexical order. Similarly, in Bakersfield, California, J. Calder and King (2022) find that cisgender heterosexual Black speakers were not significantly different in /s/ frontness with respect to gender. In this case, the Bakersfieldian Black indexical field may associate gendered /s/ pronunciations with whiteness; this meaning could also be understood to emerge at a higher indexical order, forged through the situation of the hegemonic meaning into its racial context.

The studies discussed above illustrate the ability of speakers outside of cisgender white heterosexual populations to construct alternative and subversive meaning structures for /s/ which allow an interpretation of their unique speech practices

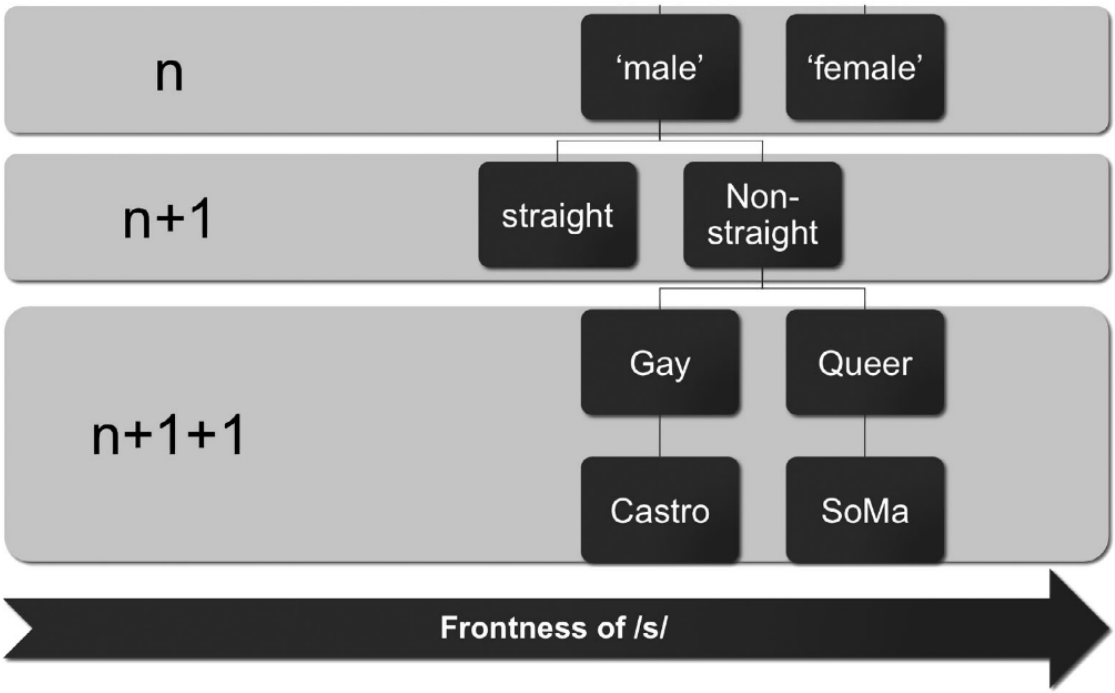


Figure 1.2: Diagram from Calder (2019a, p. 340) illustrating indexical orders of /s/ frontness and corresponding fractal recursive social oppositions across two San Fransisco drag communities.

as innovative and subversive rather than inadequate and deviant. However, due to the dominance of the hegemonic indexical field, and the practicalities of social life, it is nonetheless the case that those of us outside of cisgender white heterosexual populations have to interact with people within these populations who may inhabit the associated listening subject position and make these negative judgements of our speech. In this situation of indexical misalignment, in which speaker and listener have differing indexical inventories, it is marginalized speakers who are expected to adapt, as privileged listeners will entitle themselves to the universal application of their indexical inventory. As mentioned above, a speaker in Heffernan (2023) notes that she modifies her voice when interacting with the 'cis universe'. Drawing upon all of this prior work on indexicalities of /s/, this thesis investigates whether this sort of modulation involves /s/, and the implications and indexicalities that arise through the transfeminine speakers' behavior within the specific situation of indexical misalignment.

That individuals may change their /s/ pronunciations within interaction for gender-presentational and interactional purposes has been attested by Benesch (2022), Mendoza (2024), Saigusa (2016), Holmes-Elliott and Levon (2017), Levon and Holmes-Elliott (2013), Parnell-Mooney (2019), Calder (2019b), and Zimman (2017c). Holmes-Elliott and Levon (2017) and Levon and Holmes-Elliott (2013), in a study of within-speaker variation in /s/ pronunciation in British reality television shows, find that in one lower-class show, cisgender heterosexual women

use backer /s/ in conversation with other women, whereas in a higher-class show, cisgender heterosexual women use fronter /s/ in this context. In order to explain these contradictory results, they conclude that the interlocutor's gender may not be the best predictor, as in both shows women back /s/ in speech activities considered more threatening. As discussed below in the methods chapter, the recorded conversations I conducted attempted to control for this stance-based shifting through the structure of two collaborative-valenced speech activities.

Moving to investigate /s/ shifting in queer speakers, Calder (2019b) finds that as SoMa drag queens get into their drag, visually transforming into their drag persona, their /s/ pronunciations get progressively fronter to index their change in persona. Benesch (2022) finds that among genderfluid people, who experience different genders at different times, shifts in /s/ index these shifts in gender, however they do not necessarily correlate with typical scales of gender such as masculinity vs. femininity, but rather with personal, alternative scales such as 'flannel' vs. 'glitter'. In a case study of a Latinx drag queen part of a drag community in Wynwood, Miami who exhibited particularly wide variation in /s/ pronunciations, Mendoza (2024) finds this queen uses fronter /s/ when discussing the aspects of their drag taken from white, higher-class femininity, but uses backer /s/ when aligning themself to the more anti-normative parts of their *chonga* drag persona, where *chonga* is a queer subversive Latinx femininity which may look like "unshaved body hair and beards, as well as hoop earrings and dramatic lip liner" (307). The polyvalence of this queen's drag persona is indexed through these changes in /s/. Specifically of note here is that for them, fronted /s/ is not indexical of femininity in general, but rather indexical of a specific white cisgender heterosexual higher-class femininity. In this way, meanings are reinterpreted in order to subvert the white cisgender listening subject.

I choose to focus on /s/ rather than other lingiuistic forms bearing social indexicalities because of the theoretical depth of the research that has already been conducted on /s/; this research has captured a nuanced and detailed view of the hegemonic indexical field of /s/, how it is marginalizing, and how those who are marginalized by it reinterpret and construct its meanings subversively within their communities. However, in the /s/ literature, as well as in sociolinguistic research more broadly, a gap I address is the dearth of research looking specifically at situations of conflicting indexical fields, analyzing how interactions play out between individuals with different indexical inventories. Comparing transfeminine individuals' /s/ pronunciations in conversations with another transfem, where they can adopt their own indexicalities more freely, against their /s/ pronunciations in conversations with a cis woman, is illuminating here theoretically. More broadly, this research also addresses the absence of intra-speaker research on transfeminine speakers– although there have been of studies of intra-speaker variation in transmasculine people (eg. Zimman 2017c; Parnell-Mooney 2019), nonbinary people (Gratton 2016), and drag queens (Calder 2019b; Mendoza 2024), there have been no such studies focused on how transfeminine speakers speak differently in differing situations. A final question my study hopes to address is to understand how transfeminine individuals navigate the trans-exclusive fractal-recursive indexical

orders of /s/ indexicality, as developed by Calder (2019a) and discussed above, when forging new indexicalities may not be recognized or well-received by a cisgender listener. This thesis thus persues the following research questions:

1. How do transfeminine speakers' /s/ pronunciations differ between speech with another transfeminine person versus with a cisgender person?
2. How are transfeminine speakers able to use /s/ to index their genders and sexualities?
3. How do transfeminine speakers linguistically navigate the hegemonic semiotic structure of /s/ which is invested in denying their gender and signifying them as deficient?

1.3 extra stuff

Zimman (2013) finds that a group of cisgender, majority-heterosexual listeners rated cis gay voices and trans men's voices as similarly gay-sounding, despite differences in these groups' language use: trans men's /s/ pronunciations were significantly lower in CoG than cis gay men's and significantly higher in skewness. He concludes that in judging gay-soundingness, percievers do not listen for a single linguistic feature or set of features but instead judge deviance from cisgender hegemonic masculine speech across many linguistic features.

zimman_

Chapter 2

Methodology

In order to investigate whether and how transfeminine individuals speak differently with transgender people than they do with cisgender people, and to be able to interpret the possible motivations behind these patterns of change, I recruited four transfeminine individulals to be speakers and two women, one cisgender and one transgender, to be conversation partners. Speakers participated in an audio-recorded, semi-structured casual conversation session with the trans conversation partner, then the cis partner, then completed a demographic survey, then participated in an audio-recorded semi-structured interview with me about their identity and transgender experience. This order of study activities was selected to minimize priming effects, since it is possible questions asked in the survey and interview could effect subsequent linguistic behavior.

2.1 Participants

All participants were students of Reed College between the ages of 18 and 25, recruited by flyers, word-of-mouth, and email. All speakers identified as transfeminine, and both convesation partners identified as women; one identified as cisgender and one identified as transgender. Participants are referred to in this thesis by pseudonyms.

2.1.1 Survey

On the survey, participants were asked to provide all gender, sexuality, and race labels they identify with, as well as their pronouns. With respect to gender identity, I additionally asked participants to rank on the survey how much they generally identify with femininity and masculinity on scales of 0 to 100, acknowledging that these identifications are likely not static and subject to change throughout time (see Benesch 2022). The survey also asked transfeminine speakers questions about aspects of the body which may have an effect on speech: whether the participant was currently on hormone replacement therapy, and for how long they had been on it, whether the participant's vocal tract had elongated in puberty due to

testosterone (see Zimman 2017a; Zimman 2017c), and whether they had received voice feminization surgery. Speakers were also asked whether they had received voice feminization therapy, other forms of speech therapy, or had any disabilities related to speech. Questions were also included to understand speakers' linguistic backgrounds– what languages they speak, whether they have primarily spoken English throughout their life or have had periods of primarily speaking other languages, and the reigon(s) in which they grew up. Participants' survey responses are summarized in tables 2.1.1 and 2.1.1.

Pseudonym	Pronouns	Role	Gender	Sexuality	Race	Linguistic Background
Examplina	she/they	Speaker	Nonbinary	Queer	Latinx	English-Spanish Bilingual
...						

Table 2.1: Demographic Survey Responses [this table will be filled in once data has been collected.]

Pseudonym	Femininity (0-100)	Masculinity (1-100)	On HRT?	HRT Dur.	Voice Interventions?
Examplina	82	40	Yes	$<$1yr	Some VFT
...					

Table 2.2: Gender Background Responses (HRT: hormone replacement therapy, VFT: vocal feminization therapy: VFS: vocal feminization surgery) [this table will be filled in once data has been collected.]

2.2 Conversations

Audio-recorded, semi-structured casual conversation sessions were conducted in a sound-isolated recording booth using clip-on lavalier microphones (on the speaker as well as the conversation partner) connected to a Zoom h4n recorder. Participants were told to spend the first 10-15 minutes of the recording time collaboratively playing a word game of their choice (mini-crossword, New York Times Wordle, New York Times Connections, etc.). This prompting was intended to get participants comfortable talking with each other on the recording, and to take up a collaborative and friendly stance towards one another. Controlling for stance in this way was intended to isolate gender-related variation in /s/, as indexicalities between /s/ and confrontational/threatening stances have been found (see Holmes-Elliott and Levon (2017)). After they were finished playing, I asked participants to spend the rest of the time of the recording, another 10-15 minutes, getting to know each other, trying to find things in common, and talking about shared experiences. This portion was intended to represent more casual unstructured talk with a similarly friendly stance orientation, and in conversations with

the trans conversation partner, provide participants the opportunity to talk with each other about being transgender.

2.3 Interviews

After conversation sessions and surveys were completed, I also conducted audio-recorded semi-structured interviews with the speakers. The questions in these interviews all shared the aim of collecting background through which I am able to interpret and analyze speakers' behavior. Aside from asking follow up questions to survey responses myself and the participant were interested in elaborating on, I additionally asked questions modeled after those used by Heffernan (2023) to investigate speakers' understandings of the cisgender listening subject, such as 'do you speak differently to different types of people?' and 'Are there times or situations when you are more aware of or self-conscious about your voice?' With these questions, I was interested in the attitudes speaker participants expressed towards cisgender listeners and whether they thought they presented their voice and gender differently to cisgender people than to other transgender people. In this way, I would be able to gain access to some of the knowledge and beliefs which may have informed speakers' semiotic behavior in the recordings. This would allow me to understand these speakers' indexical inventories and their knowledge of and attitudes towards different indexical fields. I additionally asked about their close social circle in order to get a sense of any possible local nonhegemonic indexical fields the speakers may hold awareness of. I also asked questions which aimed to get a broader sense of how speakers understand how they are perceived, such as "Do you feel like people can tell you're trans, or that you get 'clocked'? If so, what do you think tips them off?" and "When you haven't told someone about your gender, or that you're trans, what do they seem to assume about your gender?" I also included in the interview questions which get at the intersections of various identities, such as "Do you feel that your voice sounds lesbian?" and "How do you think being Asian effects people's perception of your gender and your voice?", as it was important to understand whether speakers felt flattened and homogenized as transgender, or seperated by their other identities. A full list of questions is provided in appendix B. Profiles of the speakers are given in the next section based on their survey responses and interview responses.

2.4 Analytical Approach

I then conducted a within-speaker and between-speaker analysis, both of which have quantitative and qualitative portions.

For the within-speaker analysis, quantitative analysis compares speakers' /s/ pronunciations when speaking to the trans conversation partner versus when speaking to the cis conversation partner. For this analysis I adopt mean CoG of /s/, measured in Hz, as the dependent variable, and trans status of the conversation

partner (cis vs. trans) as the primary independent variable, also considering conversation module (game vs. talk). I additionally conduct some qualitative analysis on moments of note in which topic of discussion or momentary stance may have impacted /s/ production in ways relevant to my research questions.

For the between-speaker analysis, in which I investigate whether differences between transfeminine speakers impact this within-speaker behavior, comparing the ways in which different participants differently shift. For the quantitative portion of this analysis I take difference in the means of /s/ CoG between the conversation with the cisgender conversation partner and the conversation with the transgender conversation partner (to operationalize the extent and directionality of shifting) as the dependent variable, and take survey responses as independent variables. For the qualitative portion of this analysis, I also consider differences in speakers' shifting behavior based on details on speakers identities, backgrounds, and attitudes from the interviews.

In order to measure /s/ CoG, I removed the first and last 3 minutes of each recording to exclude any unnatural speech due to the unfamiliar recording setting. I then used the linguistic acoustic analysis software Praat (Boersma and Weenik 2023), to manually delimit the start and end time of each /s/ produced by each speaker in their recording, and used a script by DiCanio (2021) to measure the mean CoG of each of these /s/ tokens. I used the statistical analysis software R (R Core Team 2025) to combine, statistically analyze, and produce visuals.

In my discussion and interpretation, I employ information learned about the speakers in the survey and interview in analyzing speakers' behavior. In this way, I aim to analyze speakers on their own terms, rather than to compare them against a white cisgender heteronormative standard which will inevitably find them somehow faulty (see Calder and A. Steele 2025; Zimman 2021a). Instead, I hope to highlight the ways that speakers achieve their semiotic goals through applying their nuanced understandings of gender and the voice in their everyday linguistic practice.

Chapter 3

Participant Profiles

Chapter 4

Quantitative Results and Analysis

4.1 Within-speaker

4.2 Between-speaker

Chapter 5

Discussion and Qualitative Analysis

5.1 Comparison with Prior Results

Conclusion

5.2 Directions for Future Research

Appendix A

Survey

Appendix B

Interview Questions

Bibliography

Austin, J. L. (1962). *How to do things with words*. The William James lectures 1955. Oxford: Clarendon Press.

Barrett, Rusty and Kira Hall (Oct. 21, 2024). "Sexuality Discourses: Indexical Misrecognition and the Politics of Sex". In: *Annual Review of Anthropology* 53 (Volume 53, 2024), pp. 127–146. ISSN: 0084-6570, 1545-4290. DOI: 10.1146/annurev-anthro-052721-092531. URL: https://www.annualreviews.org/content/journals/10.1146/annurev-anthro-052721-092531 (visited on 09/13/2025).

Becker, Kara, Sameer ud Dowla Khan, and Lal Zimman (July 2022). "Beyond binary gender: creaky voice, gender, and the variationist enterprise". In: *Language Variation and Change* 34.2, pp. 215–238. ISSN: 0954-3945, 1469-8021. DOI: 10.1017/S0954394522000138. URL: https://www.cambridge.org/core/journals/language-variation-and-change/article/beyond-binary-gender-creaky-voice-gender-and-the-variationist-enterprise/63F2D76D8874D095FFBDC445FECB4532 (visited on 10/03/2025).

Benesch, Montreal (May 1, 2022). "/s/tylizing the /s/elf: A First Look into the Concurrent Fluidity of Gender and Language". PhD thesis. DOI: 10.13140/RG.2.2.21466.62405.

Boersma, Paul and David Weenik (2023). *Praat: doing phonetics by computer [Computer program]*. Version 6.3.10. URL: http://www.praat.org/.

Bucholtz, Mary (2014). "The Feminist Foundations of Language, Gender, and Sexuality Research". In: *The Handbook of Language, Gender, and Sexuality*. John Wiley & Sons, Ltd, pp. 21–47. ISBN: 978-1-118-58424-8. DOI: 10.1002/9781118584248.ch1. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118584248.ch1 (visited on 11/02/2025).

Bucholtz, Mary and Kira Hall (2004). "Theorizing Identity in Language and Sexuality Research". In: *Language in Society* 33.4, pp. 469–515. ISSN: 00474045, 14698013. URL: http://www.jstor.org/stable/4169370 (visited on 11/05/2024).

— (2005). "Identity and interaction: a sociocultural linguistic approach". In: *Discourse Studies* 7.4, pp. 585–614. ISSN: 1461-4456. URL: https://www.jstor.org/stable/24048525 (visited on 10/20/2025).

— (Jan. 31, 2010). "Locating Identity in Language". In: *Language and Identities*. Edinburgh University Press, pp. 18–28. ISBN: 978-0-7486-3578-8. URL: https://www.degruyterbrill.com/document/doi/10.1515/9780748635788-006/html (visited on 10/19/2025).

Butler, Judith (1990). *Gender Trouble: Feminism and the Subversion of Identity*. 0th ed. New York: Routledge. 272 pp. ISBN: 978-0-203-82497-9. DOI: 10.4324/9780203824979. URL: https://www.taylorfrancis.com/books/9781136783241 (visited on 04/08/2025).

Calder, J (2019a). "From Sissy to Sickening: The Indexical Landscape of /s/ in SoMa, San Francisco". In: *Journal of Linguistic Anthropology* 29.3, pp. 332–358. ISSN: 1548-1395. DOI: 10.1111/jola.12218. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/jola.12218 (visited on 04/04/2025).

— (Feb. 2019b). "The fierceness of fronted /s/: Linguistic rhematization through visual transformation". In: *Language in Society* 48.1, pp. 31–64. ISSN: 0047-4045, 1469-8013. DOI: 10.1017/S004740451800115X. URL: https://www.cambridge.org/core/journals/language-in-society/article/fierceness-of-fronted-s-linguistic-rhematization-through-visual-transformation/ADD4290AEAAA81B3474898265590BDD6 (visited on 04/11/2025).

— (Dec. 15, 2020). "From "Gay Lisp" to "Fierce Queen": The Sociophonetics of Sexuality's Most Iconic Variable". In: *The Oxford Handbook of Language and Sexuality*. Ed. by Kira Hall and Rusty Barrett. 1st ed. Oxford University Press. ISBN: 978-0-19-021292-6 978-0-19-021294-0. DOI: 10.1093/oxfordhb/9780190212926.013.49. URL: https://academic.oup.com/edited-volume/42645/chapter/358158508 (visited on 04/05/2025).

— (2021). "Whose indexical field is it?: the role of community epistemology in indexing social meaning". In: *Proceedings of the 20th Meeting of the Texas Linguistics Society*. Ed. by Caitie Coons et al. Texas: Texas Linguistics Society, pp. 39–55. URL: https://tls.ling.utexas.edu/2021/TLS_2021_Proceedings.pdf.

Calder, J and Ariana Steele (Feb. 2025). "Interrogating the cisgender listening subject in the study of trans voices". In: *Gender and Language* 18.3, pp. 216–239. ISSN: 1747-6321. DOI: 10.3138/gl-2024-18.3-0002. URL: https://utppublishing.com/doi/10.3138/gl-2024-18.3-0002 (visited on 04/06/2025).

Calder, J. and Sharese King (2022). "Whose gendered voices matter?: Race and gender in the articulation of /s/ in Bakersfield, California". In: *Journal of Sociolinguistics* 26.5, pp. 604–623. ISSN: 1467-9841. DOI: 10.1111/josl.12584. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/josl.12584 (visited on 04/05/2025).

Campbell-Kibler, Kathryn (Feb. 1, 2011). "Intersecting Variables and Perceived Sexual Orientation in Men". In: *American Speech* 86.1, pp. 52–68. ISSN: 0003-1283. DOI: 10.1215/00031283-1277510. URL: https://doi.org/10.1215/00031283-1277510 (visited on 04/05/2025).

Deleuze, Gilles and Félix Guattari (1987). "November 20, 1923: Postulates of Linguistics". In: *A thousand plateaus: capitalism and schizophrenia*. Trans. by Brian Massumi. Minneapolis, MN: University of Minnesota Press, pp. 75–110. ISBN: 978-0-8166-1401-1.

DiCanio, Christian (2021). *Spectral moments of fricative spectra script in Praat*. Version 4.0. URL: https://www.acsu.buffalo.edu/~cdicanio/scripts/Time_averaging_for_fricatives_4.0.praat.

Eckert, Penelope (2008). "Variation and the indexical field". In: *Journal of Sociolinguistics* 12.4, pp. 453–476. ISSN: 1467-9841. DOI: 10.1111/j.1467-9841.2008.00374.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9841.2008.00374.x (visited on 09/14/2025).

— (Oct. 21, 2012). "Three Waves of Variation Study: The Emergence of Meaning in the Study of Sociolinguistic Variation". In: *Annual Review of Anthropology* 41.1, pp. 87–100. ISSN: 0084-6570, 1545-4290. DOI: 10.1146/annurev-anthro-092611-145828. URL: https://www.annualreviews.org/doi/10.1146/annurev-anthro-092611-145828 (visited on 12/12/2024).

Flipsen, Peter et al. (1999). "Acoustic Characteristics of /s/ in Adolescents". In: *Journal of speech, language, and hearing research* 42.3, pp. 663–677. ISSN: 1092-4388. DOI: 10.1044/jslhr.4203.663.

Flores, Nelson and Jonathan Rosa (June 1, 2015). "Undoing Appropriateness: Raciolinguistic Ideologies and Language Diversity in Education". In: *Harvard Educational Review* 85.2, pp. 149–171. ISSN: 1943-5045, 0017-8055. DOI: 10.17763/0017-8055.85.2.149. URL: https://meridian.allenpress.com/her/article/85/2/149/32176/Undoing-Appropriateness-Raciolinguistic-Ideologies (visited on 09/09/2025).

Fuchs, Susanne and Martine Toda (Apr. 29, 2010). "Do differences in male versus female /s/ reflect biological or sociophonetic factors?" In: *Turbulent Sounds: An Interdisciplinary Guide*. Ed. by Susanne Fuchs, Martine Toda, and Marzena Zygis. De Gruyter Mouton, pp. 281–302. ISBN: 978-3-11-022658-4. URL: https://www.degruyterbrill.com/document/doi/10.1515/9783110226584.281/html (visited on 09/26/2025).

Gal, Susan and Judith T. Irvine (1995). "The Boundaries of Languages and Disciplines: How Ideologies Construct Difference". In: *Social Research* 62.4, pp. 967–1001. ISSN: 0037-783X. URL: https://www.jstor.org/stable/40971131 (visited on 12/19/2024).

— (2019). *Signs of Difference: Language and Ideology in Social Life*. Cambridge: Cambridge University Press. ISBN: 978-1-108-49189-1. DOI: 10.1017/9781108649209. URL: https://www.cambridge.org/core/books/signs-of-difference/E69813363E4CD9927C6C8E1BD2FC3011 (visited on 11/04/2025).

Gratton, Chantal (Dec. 1, 2016). "Resisting the Gender Binary: The Use of (ING) in the Construction of Non-binary Transgender Identities". In: 7. URL: https://repository.upenn.edu/handle/20.500.14332/45127 (visited on 10/03/2025).

Hazenberg, Evan Nicholas Leo (2012). "Language and identity practice : a sociolinguistic study of gender in Ottawa, Ontario". masters. Memorial University of Newfoundland. URL: https://research.library.mun.ca/2346/ (visited on 04/05/2025).

Heffernan, Dani (2023). "Constructing the "Cisgender Listening Subject": Trans-Feminine Speakers' Commentaries on Voice and Being Heard". PhD thesis. UCLA. URL: https://escholarship.org/uc/item/5568m53k (visited on 04/06/2025).

Henry, Jacob and Kira Hall (Jan. 1, 2025). "Performative Language". In: *Reference Module in Social Sciences*. Elsevier. ISBN: 978-0-443-15785-1. DOI: 10.1016/

B978-0-323-95504-1.00654-2. URL: https://www.sciencedirect.com/science/article/pii/B9780323955041006542 (visited on 09/13/2025).

Holmes-Elliott, Sophie and Erez Levon (Sept. 1, 2017). "The substance of style: Gender, social class and interactional stance in /s/-fronting in southeast England". In: *Linguistics* 55.5, pp. 1045–1072. ISSN: 1613-396X. DOI: 10.1515/ling-2017-0020. URL: https://www.degruyterbrill.com/document/doi/10.1515/ling-2017-0020/html (visited on 09/27/2025).

Inoue, Miyako (2003). "The Listening Subject of Japanese Modernity and His Auditory Double: Citing, Sighting, and Siting the Modern Japanese Woman". In: *Cultural Anthropology* 18.2, pp. 156–193. ISSN: 1548-1360. DOI: 10.1525/can.2003.18.2.156. URL: https://onlinelibrary.wiley.com/doi/abs/10.1525/can.2003.18.2.156 (visited on 09/09/2025).

— (2004). "What Does Language Remember?: Indexical Inversion and the Naturalized History of Japanese Women". In: *Journal of Linguistic Anthropology* 14.1, pp. 39–56. ISSN: 1548-1395. DOI: 10.1525/jlin.2004.14.1.39. URL: https://onlinelibrary.wiley.com/doi/abs/10.1525/jlin.2004.14.1.39 (visited on 09/09/2025).

— (2006). *Vicarious Language: Gender and Linguistic Modernity in Japan*. Asia: Local Studies / Global Themes. Berkeley, Calif: University of California Press. ISBN: 978-0-520-24584-6.

International Phonetic Association (1999). *Handbook of the International Phonetic Association: a guide to the use of the International Phonetic Alphabet*. Cambridge, U.K. ; Cambridge University Press. ISBN: 978-0-521-65236-0.

Irvine, Judith T. and Susan Gal (2000). "Language Ideology and Linguistic Differentiation". In: Kroskrity, Paul V. *Regimes of language: ideologies, polities, and identities*. School of American Research advanced seminar series. Santa Fe, NM: School of American Research Press. ISBN: 978-0-933452-61-9. URL: https://web.stanford.edu/~eckert/PDF/IrvineGal2000.pdf.

Jessee, Emmett and J Calder (Mar. 12, 2025). "The Cisgender Listening Subject in Sociolinguistic Perception: Transgender Identity Affects Sibilant Categorization in American English". In: *Journal of Sociolinguistics* n/a (n/a). ISSN: 1467-9841. DOI: 10.1111/josl.12702. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/josl.12702 (visited on 04/06/2025).

Kiesling, Scott F. (June 4, 2009). "Style as Stance: Stance as the Explanation for Patterns of Sociolinguistic Variation". In: *Stance: Sociolinguistic Perspectives*. Ed. by Alexandra Jaffe. Oxford University Press, p. 0. ISBN: 978-0-19-533164-6. DOI: 10.1093/acprof:oso/9780195331646.003.0008. URL: https://doi.org/10.1093/acprof:oso/9780195331646.003.0008 (visited on 10/16/2025).

Levon, Erez (2014). "Categories, stereotypes, and the linguistic perception of sexuality". In: *Language in Society* 43.5, pp. 539–566. ISSN: 0047-4045. URL: https://www.jstor.org/stable/43904598 (visited on 09/18/2025).

Levon, Erez and Sophie Holmes-Elliott (Oct. 17, 2013). "East End Boys and West End Girls: /s/-Fronting in Southeast England". In: 13. URL: https://repository.upenn.edu/handle/20.500.14332/44918 (visited on 09/27/2025).

Livia, Anna and Kira Hall (Nov. 20, 1997). ""It's a Girl!": Bringing Perfarmativity Back to Linguistics". In: *Queerly Phrased: Language, Gender, and Sexuality*. Ed. by Anna Livia and Kira Hall. Oxford University Press, p. 0. ISBN: 978-0-19-510470-7. DOI: 10.1093/oso/9780195104707.003.0001. URL: https://doi.org/10.1093/oso/9780195104707.003.0001 (visited on 04/08/2025).

Maegaard, Marie and Nicolai Pharao (Dec. 2015). "/s/ Variation and Perceptions of Male Sexuality in Denmark". In: *Language, Sexuality, and Power: Studies in Intersectional Sociolinguistics*. Oxford University Press. ISBN: 978-0-19-021036-6. DOI: 10.1093/acprof:oso/9780190210366.003.0005. URL: https://doi.org/10.1093/acprof:oso/9780190210366.003.0005.

Mendoza, Dozandri C (July 1, 2024). "That /s/ tiene tumbao: Chonga-fied Sibilants and the Disidentificatory Sociophonetics of Miami Latinx Drag". In: *Gender and Language* 18.3, pp. 306–327. ISSN: 1747-633X. DOI: 10.3138/gl-2024-18.3-0006. URL: https://research.ebsco.com/linkprocessor/plink?id=a135299f-2e9e-3fda-a6da-71cb9e681a11 (visited on 09/13/2025).

Munson, Benjamin, Sarah V. Jefferson, and Elizabeth C. McDonald (Apr. 1, 2006). "The influence of perceived sexual orientation on fricative identification". In: *The Journal of the Acoustical Society of America* 119.4, pp. 2427–2437. ISSN: 0001-4966. DOI: 10.1121/1.2173521. URL: https://doi.org/10.1121/1.2173521 (visited on 11/25/2025).

Parnell-Mooney, James (2019). "Tran/s/gender: assessing the effects of the social construction of gender on speech: a focus on transgender /s/ realisations". MPhil(R). University of Glasgow. DOI: 10.5525/gla.thesis.41151. URL: https://eleanor.lib.gla.ac.uk/record=b3346280 (visited on 09/26/2025).

Pharao, Nicolai et al. (Feb. 2014). "Indexical meanings of [s+] among Copenhagen youth: Social perception of a phonetic variant in different prosodic contexts". In: *Language in Society* 43.1, pp. 1–31. ISSN: 0047-4045, 1469-8013. DOI: 10.1017/S0047404513000857. URL: https://www.cambridge.org/core/journals/language-in-society/article/indexical-meanings-of-s-among-copenhagen-youth-social-perception-of-a-phonetic-variant-in-different-prosodic-contexts/76B4354FEDB55A819FADDF321620BC33 (visited on 09/24/2025).

Podesva, Robert J. (2007). "Phonation type as a stylistic variable: The use of falsetto in constructing a persona". In: *Journal of Sociolinguistics* 11.4, pp. 478–504. ISSN: 1467-9841. DOI: 10.1111/j.1467-9841.2007.00334.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9841.2007.00334.x (visited on 10/01/2025).

Podesva, Robert J. and Janneke Van Hofwegen (Dec. 24, 2015). "/s/exuality in Smalltown California: Gender Normativity and the Acoustic Realization of /s/". In: *Language, Sexuality, and Power: Studies in Intersectional Sociolinguistics*. Ed. by Erez Levon and Ronald Beline Mendes. Oxford University Press, pp. 168–188. ISBN: 978-0-19-021036-6. DOI: 10.1093/acprof:oso/9780190210366.003.0009. URL: https://doi.org/10.1093/acprof:oso/9780190210366.003.0009 (visited on 04/05/2025).

R Core Team (2025). *R: A Language and Environment for Statistical Computing*. Vienna, Austria. URL: https://www.R-project.org/.

Rickford, John R and Faye Mcnair-Knox (1994). "Addressee- And Topic-Influenced Style Shift: A Quantitative Sociolinguistic Study". In: *Sociolinguistic Perspectives On Register*, pp. 235–276. DOI: 10.1093/oso/9780195083644.003.0011.

Saigusa, Julie (Mar. 21, 2016). "Jane Lynch and /s/: The Effect of Addressee Sexuality on Fricative Realization". In: *Lifespans and Styles* 2.1, pp. 10–16. ISSN: 2057-1720. DOI: 10.2218/ls.v2i1.2016.1426. URL: https://journals.ed.ac.uk/lifespansstyles/article/view/1426 (visited on 09/16/2025).

Silverstein, Michael (July 1, 2003). "Indexical order and the dialectics of sociolinguistic life". In: *Language & Communication*. Words and Beyond: Linguistic and Semiotic Studies of Sociocultural Order 23.3, pp. 193–229. ISSN: 0271-5309. DOI: 10.1016/S0271-5309(03)00013-2. URL: https://www.sciencedirect.com/science/article/pii/S0271530903000132 (visited on 04/09/2025).

— (2023). *Language in culture: lectures on the social semiotics of language*. In collab. with E. Summerson Carr, Susan Gal, and Constantine V. Nakassis. Cambridge, United Kingdom ; Cambridge University Press. ISBN: 978-1-009-19883-7.

Steele, Ariana J. (Apr. 20, 2022). "Enacting New Worlds of Gender: Nonbinary Speakers, Racialized Gender, and Anti-Colonialism". In: *The Oxford Handbook of Language and Sexuality*. Ed. by Kira Hall and Rusty Barrett. 1st ed. Oxford University Press. ISBN: 978-0-19-021292-6 978-0-19-021294-0. DOI: 10.1093/oxfordhb/9780190212926.013.74. URL: https://academic.oup.com/edited-volume/42645/chapter/358160465 (visited on 04/05/2025).

Strand, Elizabeth A. (Mar. 1, 1999). "Uncovering the Role of Gender Stereotypes in Speech Perception". In: *Journal of Language and Social Psychology* 18.1, pp. 86–100. ISSN: 0261-927X. DOI: 10.1177/0261927X99018001006. URL: https://doi.org/10.1177/0261927X99018001006 (visited on 11/22/2025).

Zimman, Lal (Feb. 27, 2013). "Hegemonic masculinity and the variability of gay-sounding speech: The perceived sexuality of transgender men". In: *Journal of Language and Sexuality* 2.1, pp. 1–39. ISSN: 2211-3770, 2211-3789. DOI: 10.1075/jls.2.1.01zim. URL: http://www.jbe-platform.com/content/journals/10.1075/jls.2.1.01zim (visited on 04/04/2025).

— (June 2017a). "Gender as stylistic bricolage: Transmasculine voices and the relationship between fundamental frequency and /s/". In: *Language in Society* 46.3, pp. 339–370. ISSN: 0047-4045, 1469-8013. DOI: 10.1017/S0047404517000070. URL: https://www.cambridge.org/core/journals/language-in-society/article/gender-as-stylistic-bricolage-transmasculine-voices-and-the-relationship-between-fundamental-frequency-and-s/9BC20C85B373764932F69388750544A (visited on 12/02/2025).

— (2017b). "Linguistic Self-Determination and the Dialogic Nature of Identity". In: *Representing Trans: Linguistic, Legal, and Everyday Perspectives*. Ed. by Evan Hazenberg and Miriam Meyerhoff. Wellington: Victoria University Press. ISBN: 978-1-77656-175-9.

— (Sept. 1, 2017c). "Variability in /s/ among transgender speakers: Evidence for a socially grounded account of gender and sibilants". In: *Linguistics* 55.5, pp. 993–

1019. ISSN: 1613-396X. DOI: 10.1515/ling-2017-0018. URL: https://www.degruyterbrill.com/document/doi/10.1515/ling-2017-0018/html (visited on 09/11/2025).

— (2020). "Transgender Language, Transgender Moment: Toward a Trans Linguistics". In: *The Oxford Handbook of Language and Sexuality*. Ed. by Kira Hall and Rusty Barrett. Oxford University Press, p. 0. ISBN: 978-0-19-021292-6. DOI: 10.1093/oxfordhb/9780190212926.013.45. URL: https://doi.org/10.1093/oxfordhb/9780190212926.013.45 (visited on 12/19/2024).

— (Oct. 6, 2021a). "Beyond the cis gays' cis gaze: The need for a trans linguistics". In: *Gender and Language* 15.3. ISSN: 1747-633X, 1747-6321. DOI: 10.1558/genl.20883. URL: https://journal.equinoxpub.com/GL/article/view/20883 (visited on 12/19/2024).

— (2021b). "Gender diversity and the voice". In: *The Routledge Handbook of Language, Gender, and Sexuality*. Routledge, pp. 69–89. URL: https://doi.org/10.4324/9781315514857.